Model Checking of Knowing How Logics

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Abstract

TODO!

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1 How to use this?

To generate the PDF, open `report.tex` in your favorite LaTeXeditor and compile. Alternatively, you can manually do `pdflatex report; bibtex report; pdflatex report; pdflatex report` in a terminal.

You should have stack installed (see `https://haskellstack.org/`) and open a terminal in the same folder.

- To compile everything: `stack build`.
- To open ghci and play with your code: `stack ghci`
- To run the executable from Section **??**: `stack build && stack exec myprogram`
- To run the tests from Section **??**: `stack clean && stack test --coverage`

2 Introduction

Ability-based epistemic logics are found many places in computer science, as arguably they provide us with a formal approach for strategic reasoning and AI planning. Several logics in this family have been studied lately. In particular, epistemic logics describing the notion of *knowing how* received a lot of attention, due to their simplicity and good meta-logical properties.

The simplest version of *knowing how*, $\mathcal{L}_{Kh}$, based on the paper by (Wang 2015[3]), with the semantics reformulated in (Areces et al. 2021[1]). This logic captures the agent's ability to guarantee the completion of a task under a given condition. This idea is ideal in a sense, since even though the agent manages to do something under a condition, it does not imply the agent really know how. For example, one can win a lottery does not imply one can guarantee how to win. The uncertainty is involved in this case. Therefore, another logic may be prefered, $\mathcal{L}^U_{Kh}$. In this logic, an agent may have *indistinguishability* among several plans they might take, a critical concept in epistemic logic. Indistinguishable plans form equivalence classes, partitioning the set of available plans. Intuitively, plans considered identical by the agent are placed in the same class. Furthermore, this logic involves multi-agents, as opposed to a single agent. Finally, a new logic, reg-$\mathcal{L}^U_{Kh}$, a variation of the second version, introduces equivalence classes that are defined in terms of finite automatons.

In this project, we aim to model this family of knowing-how logics. We also intend to implement the model checking for the logic reg-$\mathcal{L}^U_{Kh}$ defined in [2].

Our project goals are defined as follows.

- Get familiar with the family of knowing-how logics;
- Implement the the family of knowing-how logics (syntax, semantics) in Haskell;
- Implement the explicit model checker for the basic knowing how logic;
- Implement model checking problem for the logic reg-$\mathcal{L}^U_{Kh}$ defined in [2];
- (Optional) Implement testing for model checking.
- (Optional) Implement a user-interface or at the very least an executable (using parsing) for model checking.

3 Basic Knowing How Logic $\mathcal{L}_{Kh}$ in Haskell

In this section we model the language of *basic knowing how* $\mathcal{L}_{Kh}$ introduced in [3]. This logic captures the ability of an agent to achieve a goal φ given a precondition ψ, by requiring the existence of a plan that is strongly executable and goal-reaching from every state satisfying ψ. The underlying models are labelled transition systems, where each relation corresponds to an action the agent can perform.

We implement an explicit model-checker for this logic with a bounded plan depth, along with QuickCheck generators for random testing of the semantics.

3.1 Preliminaries

Definition 3.1. (Syntax) Given a set of proposition letters $Prop$, the language $\mathcal{L}_{Kh}$ is defined by

$$\varphi ::= \top \mid p \mid \neg\varphi \mid \varphi \wedge \varphi \mid Kh(\varphi, \varphi),$$

where $p \in Prop$.

Definition 3.2 (LTS)**.** A labelled transition system (LTS) is a tuple $\mathcal{M} = (S, (R_a)_{a \in Act}, V)$, where:

- S is a non-empty set of states,
- $(R_a)_{a \in Act}$ is a family of binary relations on S, each labelled by an action $a \in Act$,
- $V : S \to 2^{Prop}$ is a valuation function.

We write $s \xrightarrow{a} t$ if $(s, t) \in R_a$. For a plan $\sigma = a_1 \dots a_n \in Act^*$, we write $s \xrightarrow{\sigma} t$ if there exist $s_1, \dots, s_{n-1}$ such that $s \xrightarrow{a_1} s_1 \xrightarrow{a_2} \cdots \xrightarrow{a_n} t$. A plan $\sigma = a_1 \dots a_n$ is *strongly executable* at s iff for every $0 \leq k < n$ and every t such that $s \xrightarrow{\sigma_k} t$, the state t has at least one a_{k+1}-successor.

Definition 3.3 (Semantics)**.** Let $\mathcal{M}$ be an LTS, and s be a state.

$\mathcal{M}, s \models \top$	$\Leftrightarrow$	*always*
$\mathcal{M}, s \models p$	$\Leftrightarrow$	$p \in V(s)$
$\mathcal{M}, s \models \neg\varphi$	$\Leftrightarrow$	$\mathcal{M}, s \not\models \varphi$
$\mathcal{M}, s \models \varphi \wedge \psi$	$\Leftrightarrow$	$\mathcal{M}, s \models \varphi$ and $\mathcal{M}, s \models \psi$
$\mathcal{M}, s \models Kh(\psi, \varphi)$	$\Leftrightarrow$	there exists a $\sigma \in Act^*$ such that for all s' such that $\mathcal{M}, s' \models \psi$: σ is strongly executable at s' and for all t such that $s' \xrightarrow{\sigma} t$, $\mathcal{M}, t \models \varphi$

The $Kh(\psi, \varphi)$ can be interpreted as "the agent knows how to achieve φ given ψ". The abbreviations $\bot$, $\vee$, $\to$, and the universal modality $U_\varphi := Kh(\neg\varphi, \bot)$ have been left out for simplicity.

3.2 Basic notions in Haskell

The syntax is modelled following Definition 3.1.

```haskell
module SingleAgent where

import Data.List (nub, delete)
-- import NoneEmpty including its constructor :|
import Data.List.NonEmpty (NonEmpty(..), toList)
```

```haskell
import Test.QuickCheck

type Proposition = Int

data Form = P Proposition | Neg Form | Conj Form Form | KH Form Form | T
    deriving (Eq, Show, Ord)
```

Following Definition 3.2, we model the LTS as follows. When creating a LTS, consider that states :: NonEmpty State, and therefore must use the :| constructor, i.e. (n :| [n+1..]). For more information see NonEmpty on Hoogle.

```haskell
type Action = Integer
type Plan = [Action]
type State = Int
type Valuation = [(State, [Proposition])]

data AbilityMap = LTS {
    states :: NonEmpty State,
    transitions :: Relations,
    valuation :: Valuation
} deriving(Show)
```

The relations and strong executability in Definition 3.2 are implemented below. Both of these are binary relations on states. Therefore, in our implementation, we use the same Haskell type `Rel` to represent them, since both can be understood as sets of pairs of states.

To represent the family of atomic relations $R = (R_a)_a$, we index relations by actions. This gives rise to the type `Relations`, corresponding to the collection of action-labelled transitions in the literature.

We do not model R_σ explicitly as a separate data structure. Instead, we only implement the operations needed to reason about plan execution and strong executability.

For this purpose, we define the following helper functions:

- `image`, which computes the image of a state under a relation, i.e. it collects all successor states of a given state;
- `r_a`, which takes an action a and the family of relations R, and returns the indexed relation R_a;
- `executePlan`, which computes all possible end states obtained after executing a plan from a given initial state;
- `stronglyExecutableAt`, which determines whether a plan is strongly executable at a given state.

```haskell
-- Both $R_a$ and $R_\sigma$ share the same type.
-- Here we model the whole set of relations, without classification.
type Rel = [(State, State)]

-- The family of action-indexed relations (R_a)_a
type Relations = [(Action, Rel)]

-- Image of a state under R_a
image :: Rel -> State -> [State]
image r u = [v | (x, v) <- r, x == u]

-- Given (R_a)_a and an action x, return R_x
r_a :: Relations -> Action -> Rel
```

```haskell
r_a rs a =
    case lookup a rs of
        Just r  -> r
        Nothing -> []

-- Execute a plan from a given start state
-- This corresponds to R_sigma(u)
executePlan :: Relations -> State -> Plan -> [State]
executePlan _  u []       = [u]
executePlan rs u (a:sigma) =
    nub (concat [ executePlan rs v sigma | v <- image (r_a rs a) u ])

-- Plan is SE or not at a state
stronglyExecutableAt :: Relations -> State -> Plan -> Bool
stronglyExecutableAt _  _ []       = True
stronglyExecutableAt rs u (a:sigma) =
    let next = image (r_a rs a) u
    in  not (null next) &&
        all (\v -> stronglyExecutableAt rs v sigma) next
```

3.3 Model checker in Haskell

We now implement the semantics in Definition 3.3. Note that `findPlans` enumerates all plans up to a fixed depth (currently 5). This is a bounded approximation: if a witness plan longer than the bound exists, the checker will not find it. A complete decision procedure would require another PSPACE-Complete algorithm in Theorem 1 from [2].

```haskell
-- Given an LTS and formula, returns the states that satisfy said formula
statesSatisifying :: AbilityMap -> Form -> [State]
statesSatisifying m f = [s | s <- toList (states m), isTrue (m, s) f]

-- Given an LTS, find all plans.
findPlans :: AbilityMap -> [Plan]
findPlans m = nub (concatMap (plansFrom depth) (toList (states m)))
  where
    -- For now we limit the depth to avoid infinite loops.
    depth = 5

    -- Flatten transitions
    edges :: [(Action,(State,State))]
    edges = [ (a,(u,v))| (a,rel) <- transitions m, (u,v) <- rel ]

    -- Generate all plans from a given state up to our specified depth
    plansFrom :: Int -> State -> [Plan]
    plansFrom 0 _ = []
    plansFrom d s =
        [ [a] | (a,(x,_)) <- edges, x == s ] -- single action plans
        ++ [ a:p | (a,(x,y)) <- edges, x == s, p <- plansFrom (d-1) y ]

isTrue :: (AbilityMap, State) -> Form -> Bool
isTrue _ T = True
isTrue (m, s) (P p) =
  case lookup s (valuation m) of
    Just props -> p `elem` props
    Nothing -> False
isTrue (m, s) (Neg f) = not (isTrue (m, s) f)
isTrue (m, s) (Conj f g) = isTrue (m, s) f && isTrue (m, s) g
-- KH is NOT local; its truth does not depend on the state at which it is evaluated.
-- KH either holds at all states, or none of them.
isTrue (m, _) (KH f g) =
    any (\a ->
        all (\s ->
            stronglyExecutableAt rs s a
            && all (\t -> isTrue (m,t) g) (executePlan rs s a)
        ) statesF
    ) candidatePlans
```

```
  where
    rs = transitions m
    statesF = statesSatisifying m f
    candidatePlans = findPlans m

-- Infix alias for the satisfaction relation
(|=) :: (AbilityMap, State) -> Form -> Bool
(|=) = isTrue
```

3.4 QuickCheck for $\mathcal{L}_{Kh}$

Finally, for this section we define the instances of Arbitrary for `Form` and `Arbitrary` respectively. For `Form` we use the `sized` function to ensure that the generated formulas remain finite in size.

```
instance Arbitrary Form where
    arbitrary = sized randomForm where
        -- Helper function to generate random formulas of a given size
        randomForm :: Int -> Gen Form
        randomForm 0 = oneof [P <$> choose (1, 5), return T]
        randomForm n = oneof
            [ P <$> choose (1, 5)
            , return T
            , Neg <$> randomForm (n - 1)
            , Conj <$> randomForm (n `div` 2) <*> randomForm (n `div` 2)
            , KH <$> randomForm (n `div` 2) <*> randomForm (n `div` 2)
            ]

instance Arbitrary AbilityMap where
    arbitrary = do
        n <- choose (1,10)
        let sts = [1..n] -- n states
        rels <- generateRelations n sts
        vals <- mapM generateValuation sts
        return $ LTS (head sts :| tail sts) rels vals
        where
            generateRelations n sts
                | n == 1 = return []
                | otherwise = do
                    m <- choose (1,n) -- decide how many actions to generate
                    actions <- vectorOf m (choose (1,5))
                    mapM (generateRel sts) actions

             -- for each action a, generate a relation labeled by a
            generateRel sts a = do
                x <- elements sts
                y <- elements (delete x sts) -- avoid loops
                return (a, [(x,y)])

            -- for each state s, generate a list of propositions
            generateValuation s = do
                props <- listOf (choose (1,5))
                return (s, nub props)
```

For now, we haven't had time yet to implement any 'interactive' elements. But it is possible to try out semantics using the definitions by running `stack ghci`. For example:

```
ghci> import Test.QuickCheck
ghci> import Data.List.NonEmpty (NonEmpty(..), toList)

ghci> generate (arbitrary:: Gen AbilityMap)
LTS {
```

```
    states = 1 :| [2,3,4,5,6],
    transitions = [(3,[(5,2)]),(4,[(2,4)]),(1,[(5,4)])],
    valuation = [(1,[4,2]),(2,[]),(3,[1]),(4,[3]),(5,[4]),(6,[])]
    }

ghci> m = ...

ghci> isTrue (m, 2::State) (KH (Conj (P 4) (Neg (P 2))) (P 3))
True

ghci> isTrue (m, 2::State) (KH (P 4) (P 1))
False
```

- HLINT ignore "Eta reduce" -

4 reg-$\mathcal{L}^U_{KH}$ in Haskell

In the framework of *basic knowing how* we introduced above, an agent possesses the ability to achieve the goal φ given ψ if and only if he has a plan that is fail-proof, meaning that each partial execution must be completable. In scenarios where the agent lacks this ability, it is only because a sequence of actions cannot be generated due to certain environmental constraints. However, another scenario may also occur: the agent does not know which plan is adequate for the situation. All he can do is blindly apply a plan he thought might work, which may or may not be successful. Such *indistinguishability* among plans establishes an epistemic relation of *knowing that*.

We now introduce a new logic of *knowing how* for a multi-agent setting defined in [2], extending the LTS with a notion of epistemic indistinguishability between plans. The uncertainty of an agent is encoded by equivalence classes of plans, and each such class is represented by a finite automaton. More precisely, each equivalence class of indistinguishable plans is represented by the language recognised by a finite automaton.

In Theorem 2 from [2], it is shown that the model-checking problem for this logic is in PTIME. The proof relies on reducing both conditions of the Kh-modality to graph reachability checks. In this section, we provide a concrete implementation of this algorithm in Haskell, closely following the constructions from the paper.

4.1 Preliminaries

Definition 4.1 (Syntax)**.** Given a set of proposition letters $Prop$ and a set of agents Agt, where $p \in Prop$ and $i \in Agt$, the language reg-$\mathcal{L}^U_{KH}$ is defined by

$$\varphi := p \mid \neg\varphi \mid \varphi \vee \varphi \mid Kh_i(\psi, \varphi).$$

We first introduce the automadta used to represent classes of plans.

Definition 4.2 (Automaton)**.** A non-deterministic finite automaton is a tuple $\mathcal{A} = (Q, Act, \delta, I, F)$, where:

- Q is a finite set of states,
- Act is a finite alphabet of actions,
- $\delta : Q \times Act \to 2^Q$ is a transition function,
- $I \subseteq Q$ is the set of initial states,
- $F \subseteq Q$ is the set of accepting (final) states.

To connect an automaton with the underlying transition system, we use the following product digraph.

Definition 4.3 (Digraph)**.** Given a reg-LTS^U $\mathcal{S}$ and an automaton $\mathcal{A} = (Q, Act, \delta, I, F)$, we define a digraph $G = (V, E)$ as follows:

- $V = Q \times S$, representing the combined states of the automaton and the underlying transition system;
- E is the set of edges such that $((q, t), (q', t')) \in E$ if and only if there exists an action $a \in Act$ such that:
  1. $q \xrightarrow{a} q'$ in $\mathcal{A}$;
  2. $(t, t') \in R_a$.

This construction synchronizes transitions of the automaton with transitions of the LTS under the same action, allowing us to track which runs of the automaton are realizable in the model.

We next recall the relational notions needed for the semantics.

Definition 4.4. For $\pi \subseteq Act^*$, $u \in S$, and $U \subseteq S$, define

$$R_\pi := \bigcup_{\sigma \in \pi} R_\sigma, \qquad R_\pi(u) := \bigcup_{\sigma \in \pi} R_\sigma(u), \qquad R_\pi(U) := \bigcup_{u \in U} R_\pi(u).$$

A set of plans $\pi \subseteq Act^*$ is *strongly executable* at $u \in S$ iff every plan $\sigma \in \pi$ is strongly executable at u. This gives rise to the definition $SE(\pi) := \bigcap_{\sigma \in \pi} SE(\sigma)$, which collects all states at which every plan in π is strongly executable.

These notions can be lifted from plan sets to automata via their accepted languages.

Definition 4.5. In terms of automata, $\mathrm{SE}(L(\mathcal{A})) := \bigcap_{\sigma \in L(\mathcal{A})} \mathrm{SE}(\sigma)$ collects all states at which every plan in $L(\mathcal{A})$ is strongly executable, and $R_{L(\mathcal{A})} := \bigcup_{\sigma \in L(\mathcal{A})} R_\sigma$ collects all transitions realisable by some plan in $L(\mathcal{A})$.

We now define reg-LTS^U models.

Definition 4.6. A reg-LTS^U is a tuple $\mathcal{S} = (S, (R_a)_{a \in Act}, (U_i)_{i \in Agt}, V)$, where the elements of U_i associated with an agent i are finite automata, namely,

$$\emptyset \neq U_i = \{\mathcal{A}_1, \mathcal{A}_2, \ldots\}.$$

Here, $L(\mathcal{A}_i)$ denotes the language accepted by $\mathcal{A}_i$. Moreover, for every $\mathcal{A}_m, \mathcal{A}_n \in U_i$ such that $m \neq n$, we have

$$L(\mathcal{A}_m) \cap L(\mathcal{A}_n) = \emptyset.$$

Finally, we can state the semantics of the language.

Definition 4.7 (Semantics)**.** Let $\mathcal{S} = (S, (R_a)_{a \in Act}, (U_i)_{i \in Agt}, V)$ be a finite reg-LTS^U, and $s \in S$. The satisfaction relation $\models$ is defined as:

$\mathcal{S}, s \models p$	*iff*	$p \in V(s)$
$\mathcal{S}, s \models \neg\varphi$	*iff*	$\mathcal{S}, s \not\models \varphi$
$\mathcal{S}, w \models \psi \vee \varphi$	*iff*	$\mathcal{S}, w \models \psi$ or $\mathcal{S}, w \models \varphi$
$\mathcal{S}, s \models Kh_i(\psi, \varphi)$	*iff*	there is a finite automaton $\mathcal{A} \in U_i$, such that (1) $[\![\psi]\!]^{\mathcal{S}} \subseteq SE(L(\mathcal{A}))$ and (2) for every $t \in [\![\psi]\!]^{\mathcal{S}}$, $R_{L(\mathcal{A})}(t) \subseteq [\![\varphi]\!]^{\mathcal{S}}$,

where $[\![\psi]\!]^{\mathcal{S}} := \{w \in S \mid \mathcal{S}, w \models \psi\}$ is the set of all states satisfying ψ.

The $Kh_i(\psi, \varphi)$ can be interpreted as "when ψ is the case, the agent i knows how to make φ true".

4.2 Basic notions in Haskell

The syntax is modelled following Definition 4.1.

```
module MultiAgent where
import SingleAgent
  ( Proposition
  , Action
  , Plan
  , State
  , Valuation
  , Relations
  , executePlan
  , stronglyExecutableAt
  , image
  , r_a
  )

import Data.List (nub)
import Data.Maybe (fromMaybe)
import GHC.Conc (par)

type Agent = Int

data RegForm = P Proposition | Neg RegForm | Disj RegForm RegForm | KH Agent RegForm RegForm
    deriving (Eq, Show, Ord)
```

The automaton type implements Definition 4.2.

```
type Successors = [State]
type ActionAtState = (State, Action)

data Automaton = ATMN {
    statesA :: [State],
    actionsA :: [Action],
    transitionsA :: [(ActionAtState, Successors)],
    initial :: [State],
    final :: [State]
} deriving (Eq, Show, Ord)
```

The product digraph from Definition 4.3 is implemented below. The helper functions `getAutNext` and `getLtsNext` retrieve successor states in the automaton and LTS respectively.

```
type GVertex = (State, State) -- (Automaton State, LTS State)
type GEdge = (GVertex, GVertex)
```

```haskell
data Digraph = Digraph {
    vSet :: [GVertex],
    eSet :: [GEdge]
} deriving (Eq, Show, Ord)

-- Helper function to get next automaton states under a given action
getAutNext :: Automaton -> State -> Action -> [State]
getAutNext atmn q a =
    fromMaybe [] (lookup (q, a) (transitionsA atmn))

-- Helper function to get next LTS states under a given action
getLtsNext :: RegLTSU -> State -> Action -> [State]
getLtsNext m t a = image (r_a (relationsM m) a) t

-- Build the product digraph G = (V, E)
buildDigraph :: RegLTSU -> Automaton -> Digraph
buildDigraph m atmn = Digraph nodes edges
  where
    nodes = [(q, s) | q <- statesA atmn, s <- statesM m]
    edges = [ ((q, t), (q', t'))
            | (q, t) <- nodes
            , act    <- actionsA atmn
            , q'     <- getAutNext atmn q act
            , t'     <- getLtsNext m t act
            ]
```

The following functions implement the relational notions from Definition 4.4.

```haskell
type PlanSet = [Plan]

-- R_pi(u) = union of R_sigma(u) for all sigma in pi
rPiU :: Relations -> State -> PlanSet -> [State]
rPiU rs u plans =
    nub (concat [executePlan rs u sigma | sigma <- plans])

-- R_pi(X) = union of R_pi(u) for all u in X
rPiX :: Relations -> [State] -> PlanSet -> [State]
rPiX rs xs plans =
    nub (concat [rPiU rs u plans | u <- xs])

-- SE(sigma) = set of all states at which sigma is strongly executable
seSigma :: [State] -> Relations -> Plan -> [State]
seSigma sts rs sigma =
    [u | u <- sts, stronglyExecutableAt rs u sigma]

-- SE(pi) = intersection of SE(sigma) for all sigma in pi
sePi :: [State] -> Relations -> PlanSet -> [State]
sePi sts rs plans =
    [u | u <- sts, all (stronglyExecutableAt rs u) plans]
```

The functions above are not used directly in the model-checking algorithm, but are included to align with the notions from the paper. For the same reason, we omit the implementation of definition 4.5. Because 4.4 and 4.5 are simply same concepts but in different perspectives.

Finally, the reg-LTS^U model implements Definition 4.6.

```haskell
type Uncertainty = [(Agent, [Automaton])] -- U_i = {A_1,...}, for each agent i

data RegLTSU = RegLTSU{
        statesM :: [State],
        relationsM :: Relations,
        uncertainty :: Uncertainty,
        valuationM :: Valuation -- use the Valuation from singleAgent
    } deriving (Eq, Show, Ord)

-- Given an agent i, return U_i = [A_1,...]
getAgentAuts :: RegLTSU -> Agent -> [Automaton]
```

```
getAgentAuts m agent =
    fromMaybe [] (lookup agent (uncertainty m))
```

4.3 Model checker in Haskell

We now discuss the implementation model checking algorithms for $reg\text{-}\mathcal{L}^U_{KH}$ in Haskell.

Given a finite reg-LTS^U $\mathcal{S} = (S, (R_a)_{a \in Act}, (U_i)_{i \in Agt}, V)$, a state $s \in S$, and a formula $Kh_i(\varphi, \psi)$, we check whether $\mathcal{S}, s \models Kh_i(\varphi, \psi)$ by iterating over all automata $\mathcal{A} \in U_a$ and verifying two conditions:

1. $[\![\varphi]\!]^{\mathcal{S}} \subseteq \mathrm{SE}(L(\mathcal{A}))$
2. $R_{L(\mathcal{A})}([\![\varphi]\!]^{\mathcal{S}}) \subseteq [\![\psi]\!]^{\mathcal{S}}$

Once we find such automaton, the formula $Kh_i(\varphi, \psi)$ is then satisfied. Since $L(\mathcal{A})$ may be infinite, directly enumerating all plans is not feasible. Following [2], we handle each condition via a separate algorithm, both of which reduce the problem to reachability checks on finite graphs.

To perform the reachability checks in PTIME, we use the following *depth-first search* algorithm:

Require: Digraph $G = (V, E)$, start vertex v_0, target set $T_0 \subseteq V$
DFS($G, v_0, T_0, \emptyset$)
function DFS($G, v, T, visited$)
    if $v \in T$ **then return true**
    end if
    if $v \in visited$ **then return false**
    end if
    $visited \leftarrow visited \cup \{v\}$
    for $w \in$ successors(G, v) **do**
        if DFS($G, w, T, visited$) **then return true**
        end if
    end for
    return false
end function

The algorithm above runs in $O(|V| + |E|)$, which is polynomial. In Haskell:

```
-- Successors of a vertex in the digraph
successors :: Digraph -> GVertex -> [GVertex]
successors g v = [v' | (u, v') <- eSet g, u == v]

-- DFS reachability: is any vertex in targetSet reachable from start?
-- The type can be read as: digraph -> start -> targets -> visited_set -> Bool
dfsReachable :: Digraph -> GVertex -> [GVertex] -> [GVertex] -> Bool
dfsReachable _ start targetSet _
    | start `elem` targetSet = True
dfsReachable _ start _ visited
    | start `elem` visited = False
dfsReachable g start targetSet visited =
    any (\v -> dfsReachable g v targetSet (start : visited)) (successors g start)
```

The algorithm that checks Condition (1) from [2] works as follows:

Require: reg-LTS^U $\mathcal{S}$, state $s \in S$, automaton $\mathcal{A} = (Q, Act, \delta, I, F)$

```
function CheckSE-Original(𝒮, s, 𝒜)
    Build the product digraph G = (Q × S, E)
    for a ∈ Act do
        for t ∈ S such that R_a(t) = ∅ do
            for q ∈ Q such that δ(q, a) ≠ ∅ do
                for q_0 ∈ I do
                    if (q, t) is reachable from (q_0, s) in G then
                        return true
                    end if
                end for
            end for
        end for
    end for
    return false
end function
```

Instead of using the nested for-loops, we first collect all *bad vertices* via set comprehension, i.e.

$$Bad = \{(q,t) \in Q \times S \mid \exists\, a \in Act : \delta(q,a) \neq \emptyset \text{ and } R_a(t) = \emptyset\},$$

and then perform a single DFS reachability check from each initial vertex (q_0, s) to the set Bad. This is equivalent to the original procedure, but avoids repeated reachability checks.

A bad vertex (q,t) represents a situation where the automaton expects to perform some action a at state q, but the LTS cannot execute a at state t. If such a vertex is reachable, it means that some plan in $L(\mathcal{A})$ fails to be strongly executable.

Remark: This function might be more accurately called `checkNotSE` *rather than* `checkSE`*, since it returns true exactly when strong executability fails.*

```
Require: reg-LTS^U 𝒮 = (S, (R_a), (U_i), V), state s ∈ S, automaton 𝒜 = (Q, Act, δ, I, F)
function CheckSE(𝒮, s, 𝒜)
    Build product digraph G = (Q × S, E)
    Bad ← {(q, t) ∈ V | ∃ a ∈ Act : δ(q, a) ≠ ∅ and R_a(t) = ∅}
    for q_0 ∈ I do
        if DFS((q_0, s)) with target Bad then return true
        end if
    end for
    return false
end function
Require: reg-LTS^U 𝒮, automaton 𝒜, truth set [[φ]]^𝒮
function CheckCond1(𝒮, 𝒜, [[φ]]^𝒮)
    for s ∈ [[φ]]^𝒮 do
        if CheckSE(𝒮, s, 𝒜) then return false
        end if
    end for
    return true
end function
```

In Haskell:

```
-- Compute the set of bad vertices in G.
-- A vertex (q, t) is bad iff there exists an action a such that:
--   (i)  the automaton has a transition from q under a, and
```

```haskell
--   (ii) the LTS has no successor from t under a.
badVertices :: RegLTSU -> Automaton -> [GVertex]
badVertices m atmn =
    nub [ (q, t)
        | q <- statesA atmn,
        t <- statesM m,
        a <- actionsA atmn,
        not (null (getAutNext atmn q a)),  --  delta(q,a) != empty
        null (getLtsNext m t a)            --  R_a(t) == empty
        ]

-- checker whether there is a path from (q0, s) to a bad vertex.
-- If so, return True. This means that s not in SE(L(A)).
-- Else, return False. This means that s in SE(L(A)).
checkSE :: RegLTSU -> State -> Automaton -> Bool
checkSE m s atmn =
    let g   = buildDigraph m atmn
        bad = badVertices m atmn
    in  any (\q0 -> dfsReachable g (q0, s) bad []) (initial atmn)

checkCond1 :: RegLTSU -> Automaton -> [State] -> Bool
checkCond1 m atmn =
    all (\s -> not (checkSE m s atmn))
```

Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be two automata. To check whether $L(\mathcal{A}_1) \cap L(\mathcal{A}_2) \neq \emptyset$, it suffices to construct the product automaton and check whether any accepting state pair in $F_1 \times F_2$ is reachable from some initial state pair in $I_1 \times I_2$.

Require: Automata $\mathcal{A}_1 = (Q_1, Act_1, \delta_1, I_1, F_1)$ and $\mathcal{A}_2 = (Q_2, Act_2, \delta_2, I_2, F_2)$

function INTERSECTIONNONEMPTY($\mathcal{A}_1, \mathcal{A}_2$)
//Build product automaton
  $V \leftarrow Q_1 \times Q_2$
  $E \leftarrow \{((q_1, q_2), (q_1', q_2')) \mid a \in Act_1 \cap Act_2,\ q_1 \xrightarrow{a} q_1' \in \delta_1,\ q_2 \xrightarrow{a} q_2' \in \delta_2\}$
  $G \leftarrow (V, E)$
  $T \leftarrow F_1 \times F_2$
  for $(q_1, q_2) \in I_1 \times I_2$ **do**
    if DFS((q_1, q_2)) with target T **then return true**
    end if
  end for
  return false
end function

To check Condition (2), it suffices to check whether

$$L(\mathcal{A}) \cap \bigcup \{L(\mathcal{A}_{(t_1,t_2)}) \mid t_1 \in [\![\varphi]\!]^{\mathcal{S}},\ t_2 \in [\![\neg\psi]\!]^{\mathcal{S}}\} = \emptyset,$$

where $\mathcal{A}_{(t_1,t_2)} = (S, Act, \delta', \{t_1\}, \{t_2\})$ is the path automaton whose transitions mirror the LTS, i.e. $\delta'(t, a) = \{t' \mid (t, t') \in R_a\}$. Its language is precisely $L(\mathcal{A}_{(t_1,t_2)}) = \{\sigma \in Act^* \mid t_2 \in R_\sigma(t_1)\}$.

By distributivity of $\cap$ over $\bigcup$, this is equivalent to checking that

$$L(\mathcal{A}) \cap L(\mathcal{A}_{(t_1,t_2)}) = \emptyset \quad \text{for all } (t_1, t_2) \in [\![\varphi]\!]^{\mathcal{S}} \times [\![\neg\psi]\!]^{\mathcal{S}}.$$

Require: reg-LTS^U $\mathcal{S}$, automaton $\mathcal{A}$, truth sets $[\![\varphi]\!]^{\mathcal{S}}$ and $[\![\neg\psi]\!]^{\mathcal{S}}$

function CHECKCOND2($\mathcal{S}, \mathcal{A}, [\![\varphi]\!]^{\mathcal{S}}, [\![\neg\psi]\!]^{\mathcal{S}}$)
  for $t_1 \in [\![\varphi]\!]^{\mathcal{S}}$ **do**
    for $t_2 \in [\![\neg\psi]\!]^{\mathcal{S}}$ **do**
      if INTERSECTIONNONEMPTY($\mathcal{A}, \mathcal{A}_{(t_1,t_2)}$) **then return false**
      end if

```
        end for
    end for
    return true
end function
```

In Haskell:

```haskell
-- Construct A_(t1, t2)
-- This automaton accepts all plans from t1 to t2
buildPathAutomaton :: RegLTSU -> State -> State -> Automaton
buildPathAutomaton m t1 t2 = ATMN
    { statesA      = statesM m,
    actionsA     = acts,
    transitionsA = [ ((s, a), getLtsNext m s a) | s <- statesM m, a <- acts],
    initial      = [t1],
    final        = [t2]
    }
  where
    -- Collect all actions that appear in the LTS relations
    acts = nub [a | (a, _) <- relationsM m]

-- Check the intersection of L(A1) and L(A2) is empty or not
intersectionNonEmpty :: Automaton -> Automaton -> Bool
intersectionNonEmpty a1 a2 =
    any (\sv -> dfsReachable prodGraph sv targets []) startVs
  where
    -- Shared actions
    sharedActs = nub [a | a <- actionsA a1, a `elem` actionsA a2]

    -- Product graph vertices and edges
    prodNodes = [(q1, q2) | q1 <- statesA a1, q2 <- statesA a2]
    prodEdges = [ ((q1, q2), (q1', q2'))
                | (q1, q2) <- prodNodes,
                a          <- sharedActs,
                q1'        <- getAutNext a1 q1 a,
                q2'        <- getAutNext a2 q2 a
                ]
    prodGraph = Digraph prodNodes prodEdges

    -- Start from any pair of initial states
    startVs = [(q1, q2) | q1 <- initial a1, q2 <- initial a2]

    -- Target: any pair of accepting states
    targets = [(q1, q2) | q1 <- final a1, q2 <- final a2]

-- Check for each A: for all (t1,t2): L(A) \cap L(A_(t1, t2)) = empty?
-- We check the emptiness of the intersection by checking the reachability of the product graph(automaton)
-- If empty, then cond 2 is SAT
checkCond2 :: RegLTSU -> Automaton -> [State] -> [State] -> Bool
checkCond2 m aut phiStates negPsiStates = not (any violates pairs)
  where
    pairs = [(t1, t2) | t1 <- phiStates, t2 <- negPsiStates]
    -- Check the intersection. Non-empty => cond 2 is UNSAT.
    violates (t1, t2) =
        let pathAut = buildPathAutomaton m t1 t2
        in  intersectionNonEmpty aut pathAut
```

Following the semantics defined in Definition 4.7 and putting things together, we complete the model checker for $reg\text{-}\mathcal{L}^{U}_{KH}$.

```haskell
-- [[phi]]= set of states that phi holds
truthSet :: RegLTSU -> RegForm -> [State]
truthSet m f = [s | s <- statesM m, isTrueReg (m, s) f]

-- Satisfaction relation for the propositional fragment
isTrueReg :: (RegLTSU, State) -> RegForm -> Bool
isTrueReg (m, s) (P p) =
    case lookup s (valuationM m) of
```

```
        Just props -> p `elem` props
        Nothing -> False
isTrueReg (m, s) (Neg f) =
    not (isTrueReg (m, s) f)
isTrueReg (m, s) (Disj f g) =
    isTrueReg (m, s) f || isTrueReg (m, s) g

-- Kh_a(phi, psi) holds iff there exists A in U_a such that
-- (1) [[phi]] is subset of SE(L(A))
-- (2) R_{L(A)}([[phi]]) is subset of [[psi]]
isTrueReg (m, _) (KH agent phi psi) =
    any (\aut -> checkCond1 m aut phiStates
              && checkCond2 m aut phiStates negPsiStates
    ) (getAgentAuts m agent)
  where
    phiStates    = truthSet m phi          -- [[phi]]
    negPsiStates = truthSet m (Neg psi)    -- [[neg psi]]

-- Infix alias for the satisfaction relation
(||=) :: (RegLTSU, State) -> RegForm -> Bool
(||=) = isTrueReg
```

5 Tests for Semantics

We now use the library QuickCheck to randomly generate models and states to verify the fundamental logical facts within the boolean fragment of our implementation.

```
module Main where

import Test.QuickCheck
import Data.List.NonEmpty (toList)
import SingleAgent
import Test.Hspec
```

The following uses the HSpec library to define our test suite. These tests ensure that our model checker respects classical propositional identities. To perform these tests, we first define an auxiliary type `PointedModel` and an `Arbitrary` instance to pick a random state from a generated model.

```
newtype PointedModel = PM (AbilityMap, State) deriving (Show)

instance Arbitrary PointedModel where
    arbitrary = do
        m <- arbitrary
        s <- elements (toList (states m))
        return (PM (m, s))
```

We verify the following principles which should hold for any boolean formula φ:

- **Double Negation Elimination**
  - $\neg\neg\varphi \equiv \varphi$
- **Commutativity of Conjunction**
  - $\varphi \wedge \psi \equiv \psi \wedge \varphi$
- **Identity of Conjunction**
  - $\varphi \wedge \top \equiv \varphi$

```haskell
-- Since Kh is currently undefined, we use a helper function to filter out formulas containing the Kh modality.
isBoolean :: Form -> Bool
isBoolean (P _)        = True
isBoolean T            = True
isBoolean (Neg f)      = isBoolean f
isBoolean (Conj f1 f2) = isBoolean f1 && isBoolean f2
isBoolean (KH _ _)     = False

-- Check if two formulas are equivalent in a specific model and state
isEquivalent :: (AbilityMap, State) -> Form -> Form -> Bool
isEquivalent (m, s) f1 f2 = isTrue (m, s) f1 == isTrue (m, s) f2

main :: IO ()
main = hspec $ do
  describe "Boolean Semantics" $ do
    it "Double Negation Elimination: Neg (Neg f) should be equivalent to f" $
      property $ \(PM (m, s)) f ->
        isBoolean f ==> isEquivalent (m, s) (Neg (Neg f)) f

    it "Commutativity of Conjunction: f1 ^ f2 should be equivalent to f2 ^ f1" $
      property $ \(PM (m, s)) f1 f2 ->
        (isBoolean f1 && isBoolean f2) ==> isEquivalent (m, s) (Conj f1 f2) (Conj f2 f1)

    it "Identity of Conjunction: f ^ T should be equivalent to f" $
      property $ \(PM (m, s)) f ->
        isBoolean f ==> isTrue (m, s) (Conj f T) === isTrue (m, s) f
```

6 Conclusion

Finally,

References

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[3] Y. Wang. A logic of knowing how. In *Logical Reasoning and Interaction (LORI'15)*, volume 9394 of *Lecture Notes in Computer Science (LNCS)*, pages 392–405. Springer, 2015.